

55 | OVERVIEW



Amplifying Heat: How Water Vapour Drives Climate Feedback

Abstract

Water vapour is Earth's most important greenhouse gas, yet it is not what is driving today's warming. In 2023, climate scientist Kevin Trenberth explained the water vapour feedback: carbon dioxide and other long-lived gases warm the atmosphere, warmer air holds more water vapour, and that extra vapour traps more heat, amplifying warming and making extreme weather worse.

Introduction

2023 brought a string of record-breaking disasters, from deadly wildfires and catastrophic floods to record-high ocean temperatures and record-low Antarctic sea ice. It seemed like climate change was accelerating, and in part it was. Since the late 1800s, global surface temperatures have risen about 1.1°C, driven by burning fossil fuels. But alongside rising emissions, scientists were seeing a long-predicted process: the water vapour feedback. Some people mistakenly believe water vapour is the cause of current warming. In fact, its rise is a consequence of warming caused by other greenhouse gases, and it acts as an amplifier.

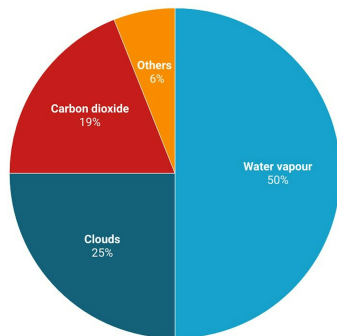
How Do Scientists Study This?

Trenberth explained the feedback using the physics of the greenhouse effect and the water cycle. Any molecule with three or more atoms can absorb and re-emit infrared radiation, giving it a blanketing effect. Scientists estimate each gas's share of the natural greenhouse effect: water vapour about 50%, clouds about 25%, carbon dioxide 19%, ozone 4%, and other gases 3%. They compare the gases' lifetimes: water vapour lasts about nine days before falling as rain or snow, while carbon dioxide stays for centuries to millennia. They measure water vapour directly and use the rule that air holds about 7% more moisture per 1°C of warming.

The main components of the atmospheric greenhouse effect

For all-sky conditions, including clouds

■ Water vapour ■ Clouds ■ Carbon dioxide ■ Others

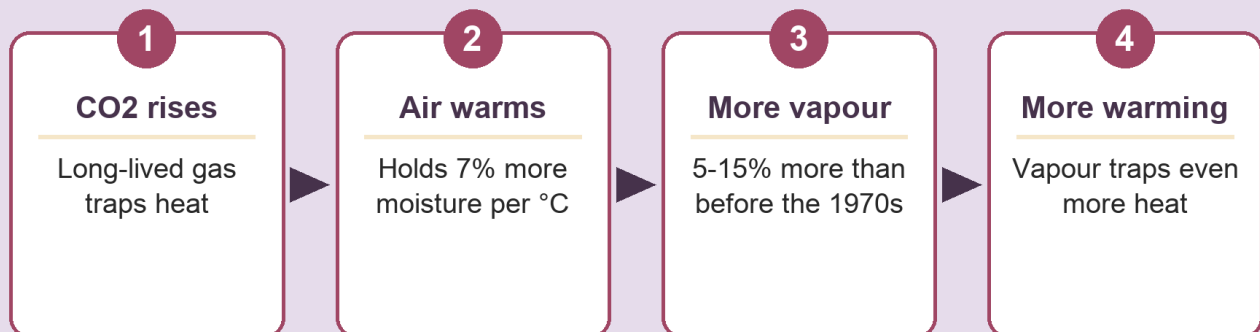


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The main components of the atmospheric greenhouse effect for all-sky conditions. Adapted from Trenberth (2022), CC BY-SA.

The water vapour feedback loop



Study the Process

Rank the gases

Estimate each gas's share of Earth's natural greenhouse effect.

Compare lifetimes

Water vapour lasts days; carbon dioxide lasts centuries or longer.

Measure the feedback

Track how much water vapour has risen as the atmosphere warmed.

What Did They Discover?

Because water vapour condenses and falls out within days, it cannot build up on its own. Long-lived gases are the backbone of atmospheric heating: carbon dioxide has risen 50%, from 280 to 420 ppm, with about half of that since 1985, and accounts for about 75% of human-caused heating from long-lived gases. As the air warms, it holds about 7% more moisture per degree Celsius. Atmospheric moisture is now an estimated 5% to 15% higher than before the 1970s, and since then, the extra heating from water vapour has been comparable to that from rising carbon dioxide. More water vapour also means heavier downpours where it rains and more drought and wildfire risk where it does not.

Water vapour by the numbers

50%

of Earth's greenhouse effect from water vapour

~9 days

average time water vapour stays in the air

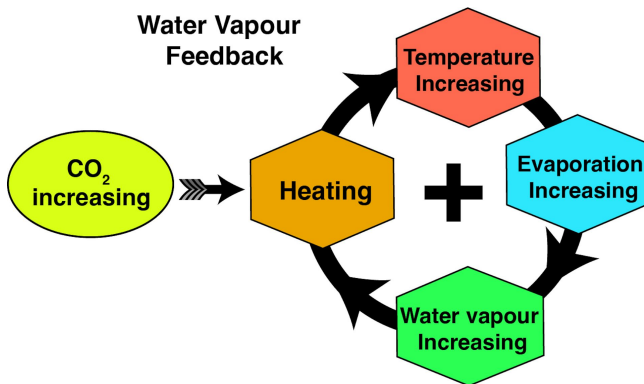
7%/°C

more moisture air can hold per degree of warming

Water is the planet's air conditioner: evaporation cools the surface, and rain washes pollution from the air, but a moister atmosphere makes storms more intense.

Look at the Evidence

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The water vapour feedback: increased heating promotes evaporation and higher temperatures, which raise water vapour levels. Author provided, CC BY-SA.

Water vapour is part of Earth's hydrological cycle. Author provided, CC BY-SA.

Takeaways

An amplifier, not a driver

Water vapour rises because of warming caused by CO₂ and other long-lived gases.

Powerful feedback

Since the 1970s, water vapour's added heating has rivaled carbon dioxide's.

Where it rains, it pours

More vapour means heavier downpours, while dry places face more drought.

Key Terms

Water vapour feedback

A cycle in which warming increases water vapour, which traps more heat and causes further warming.

Greenhouse gas

A gas, usually with three or more atoms, that absorbs and re-emits infrared heat, warming the planet.

Relative humidity

How much water vapour the air holds compared with the most it could hold at that temperature.

Evapotranspiration

The combined movement of water into the air from evaporation and from plants' leaves.

Atmospheric lifetime

How long a gas typically stays in the atmosphere before being removed.

Words to Know · Amplifying Heat

Water vapour feedback

Greenhouse gas

Relative humidity

Evapotranspiration

Atmospheric lifetime

Activity

Feedback Loop Model

Students model the water vapour feedback loop and compare greenhouse gas lifetimes.

Materials: Arrow cards; Greenhouse gas lifetime cards; Timeline strip marked in days, years, and centuries

1. Arrange cards into the feedback loop: CO₂ increasing → heating → evaporation increasing → water vapour increasing → temperature increasing.
2. Explain why the loop cannot start without CO₂ or another long-lived gas.
3. Place each gas on the timeline: water vapour (~9 days), methane (a decade or two), nitrous oxide (~a century), CO₂ (centuries to millennia).
4. Use the 7% per °C rule to calculate the extra moisture after 1°C and 2°C of warming.
5. Write a response to someone who claims water vapour, not CO₂, is causing global warming.

References

Trenberth, K. (2023). How rising water vapour in the atmosphere is amplifying warming and making extreme weather worse. The Conversation. <https://doi.org/10.64628/AA.3yf9fkrpg>